

Disinfection Byproducts Precursors Rule Regulation 11.24

Regulatory Requirement Regulation 11.24

1. For each conventional filtration treatment plant, the supplier must collect one routine paired TOC sample set and one routine source water alkalinity sample each month.
2. Determine the monthly removal ratio OR determine $SUVA < 2.0 \text{ L/mg-m}$
3. Calculate your LRAA for each Month

Violation occurs:

If the supplier fails to comply with any of the alternative compliance criteria and the LRAA of the monthly removal ratios is less than 1.00 as calculated for Step 1 or Step 2 TOC removal requirements, a disinfection byproduct precursors treatment technique violation occurs.

Step 1 TOC Removal Requirements

Regulation 11, section 11.24(5)

For each conventional filtration treatment plant, the supplier must collect one routine paired TOC sample set and one routine source water alkalinity sample each month.

- (i) The source water alkalinity sample must be collected at the same time as the paired TOC sample set and must be representative of influent water quality.

TABLE 11.24-I STEP 1 TOC PERCENT REMOVAL REQUIREMENTS			
	Source water alkalinity, mg/L as CaCO ₃		
	0-60	>60-120	>120
Source water TOC, mg/L	Required step 1 TOC percent removal		
>2.0-4.0	35.0	25.0	15.0
>4.0-8.0	45.0	35.0	25.0
>8.0	50.0	40.0	30.0

Disinfection Byproducts Rule Regulation 11.25

Regulatory Requirement Regulation

11.25

Sampling and MCL requirements for TTHM and HAA5 compounds

Quarterly dual samples on treated water sites, number of sites determined by population and source water type (Table 11.25 II)

MCL is based on LRAA per stage 2 DBPR requirements, Violation:

- LRAA at any sample point in >MCL for TTHM and/or HAA5
- LRAA (before 4 consecutive quarters) at any sampling point >MCL or TTHM and/or HAA5, regardless of subsequent sample results.

Violation:

- Notify CDPHE within 48 hours
- Tier 2 public notice

Disinfection Byproduct (DBP) formation

Form when natural organic matter in water reacts with disinfectants during treatment

Common problem for drinking water systems that use surface water due to increased organic matter

Some compounds have been deemed probably carcinogenic & possibly carcinogenic; studies show increased potential for certain cancers and adverse developmental and reproductive effects during pregnancy.

The [World Health Organization](#) has stated that "the risk of death from pathogens is at least 100 to 1000 times greater than the risk of cancer from disinfection by-products (DBPs)" {and} the "risk of illness from pathogens is at least 10 000 to 1 million times greater than the risk of cancer from DBPs".

Denver Water system- lower risk system

- Chloramination reduces formation potential, less free chlorine.
- Low organic carbon concentrations in our raw water

MCLs and sampling

TABLE 11.25-I MCLs FOR TTHM AND HAA5	
Disinfection byproduct	MCL (mg/L)
Total trihalomethanes (TTHM)	0.080
Haloacetic acids (five) (HAA5)	0.060

MCLs apply to the sum of compounds, not the individual compounds themselves.

MCLGs do exist for some individual compounds

TABLE 11.25-II ROUTINE SAMPLING LOCATIONS AND FREQUENCIES

Source water type	Population supplied	Sampling frequency	Number of sampling locations and sample type
Surface Water	<500	annually	1 individual sample for TTHM collected at the sampling location with the highest TTHM concentration and 1 individual sample for HAA5 collected at the sampling location with the highest HAA5 concentration; or 1 dual sample set if the highest TTHM and HAA5 concentrations occur at the same sampling location and in the same month.
Surface Water	500 – 3,300	quarterly	1 individual sample for TTHM collected at the sampling location with the highest TTHM concentration and 1 individual sample for HAA5 collected at the sampling location with the highest HAA5 concentration.

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Total Trihalomethanes (TTHM) and Haloacetic Acids (HAA5)

Surface Water	3,301 – 9,999	quarterly	2 dual sample sets at sampling locations as specified in 11.25(1)(c)(ii) or 11.25(1)(c)(iii).
Surface Water	10,000 – 49,999	quarterly	4 dual sample sets at sampling locations as specified in 11.25(1)(c)(ii) or 11.25(1)(c)(iii).
Surface Water	50,000 – 249,999	quarterly	8 dual sample sets at sampling locations as specified in 11.25(1)(c)(ii) or 11.25(1)(c)(iii).
Surface Water	250,000 – 999,999	Quarterly	12 dual sample sets at sampling locations as specified in 11.25(1)(c)(ii) or 11.25(1)(c)(iii).
Surface Water	1,000,000 – 4,999,999	quarterly	16 dual sample sets at sampling locations as specified in 11.25(1)(c)(ii) or 11.25(1)(c)(iii).
Surface Water	≥ 5,000,000	quarterly	20 dual sample sets at sampling locations as specified in 11.25(1)(c)(ii) or 11.25(1)(c)(iii).

disclaimer

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